TABLEAU PORTFOLIO PROJECT

GROUP A: STARTUP PERFORMANCE ANALYTICS

Startup Quadrant Analysis

Dynamic filtering, treemap sizing, and dual-encoding scatter plots

Tableau | Scatter Plot | Treemap | Bar Chart | Sets | Parameters | Dynamic Filtering

Project Summary

A 4-view interactive Tableau workbook analyzing startup performance using industry employee benchmarks, dynamic top-N treemaps, and quadrant scatter plots with dual encoding. Demonstrates LOD-adjacent set logic, parameter-driven dynamic filtering, and revenue/expense quadrant classification for identifying high-growth target startups.

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1 PROJECT OVERVIEW

Workbook:	Startup Quadrant Analysis
Views:	4 (Industries bar, Dynamic Treemap, Startup Quadrant, Scatter Plot)
Data Source:	Startup dataset -- revenue, expenses, employees, growth % by industry
Key Parameters:	Revenue Cutoff ($8.1M), Expenses Cutoff ($4.3M), Growth Leaders (Top 16)
Chart Types:	Horizontal bar chart, Treemap, Quadrant scatter plot (x2 variants)
Key Techniques:	Tableau Sets, Parameters, Dual mark encoding (shape + color), Reference lines
Business Use Case:	Identify high-growth startups within target revenue/expense quadrant

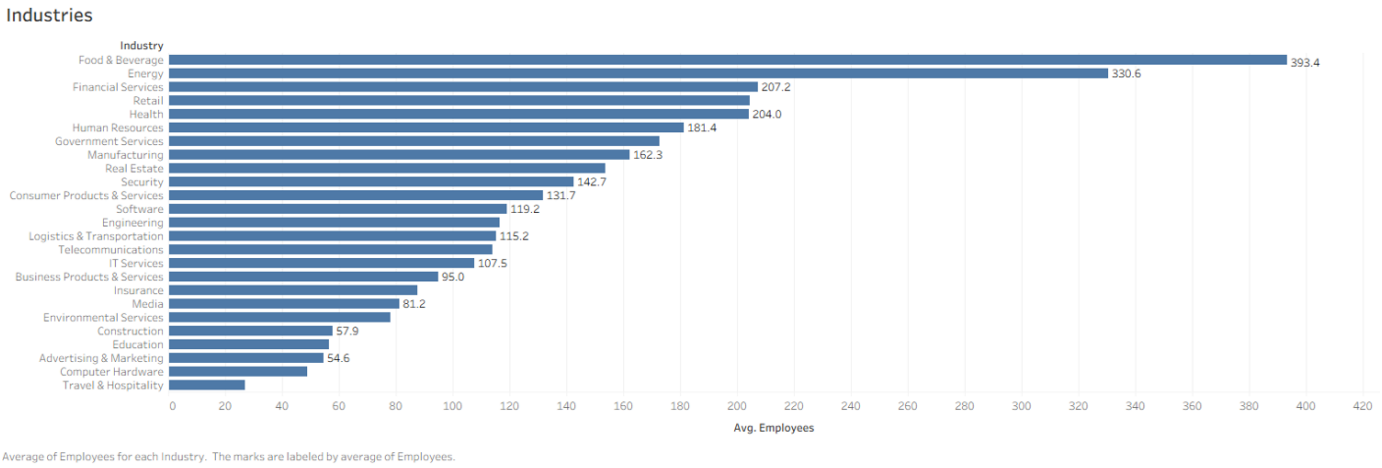
2 BUSINESS CONTEXT

Investors and accelerators need to rapidly identify startups that are both financially healthy (high revenue, controlled expenses) and exhibiting exceptional growth. This workbook solves that by combining three complementary views into a single analysis: industry benchmarks for context, a dynamic treemap for growth ranking, and a quadrant scatter for the critical revenue-vs-expense targeting decision.

- Industries bar: Shows average employee count by sector -- a proxy for startup maturity and capital intensity. Food & Beverage leads at 393.4 avg employees, Travel & Hospitality trails at ~20.
- Dynamic Treemap: Top startups ranked by 2015 growth % -- driven by a "Top N by Growth%" parameter. Rankfind leads at 784%, Matelectronics at 776%, down to Zun-Ex at 622%.
- Startup Quadrant scatter: Revenue ($8.1M cutoff) x Expenses ($4.3M cutoff) reference lines create 4 quadrants. Top 16 growth leaders highlighted in red diamonds; target quadrant (high revenue, low expenses) shown in blue.
- Scatter variant: Adds shape encoding (diamond = In top growth, circle = Out) and 4-color quadrant classification for richer cross-dimensional filtering.

3 VIEW 1 -- INDUSTRIES: AVERAGE EMPLOYEES BY SECTOR

Horizontal bar chart showing average employee headcount per industry. Used to contextualize startup size expectations before applying growth and financial filters.

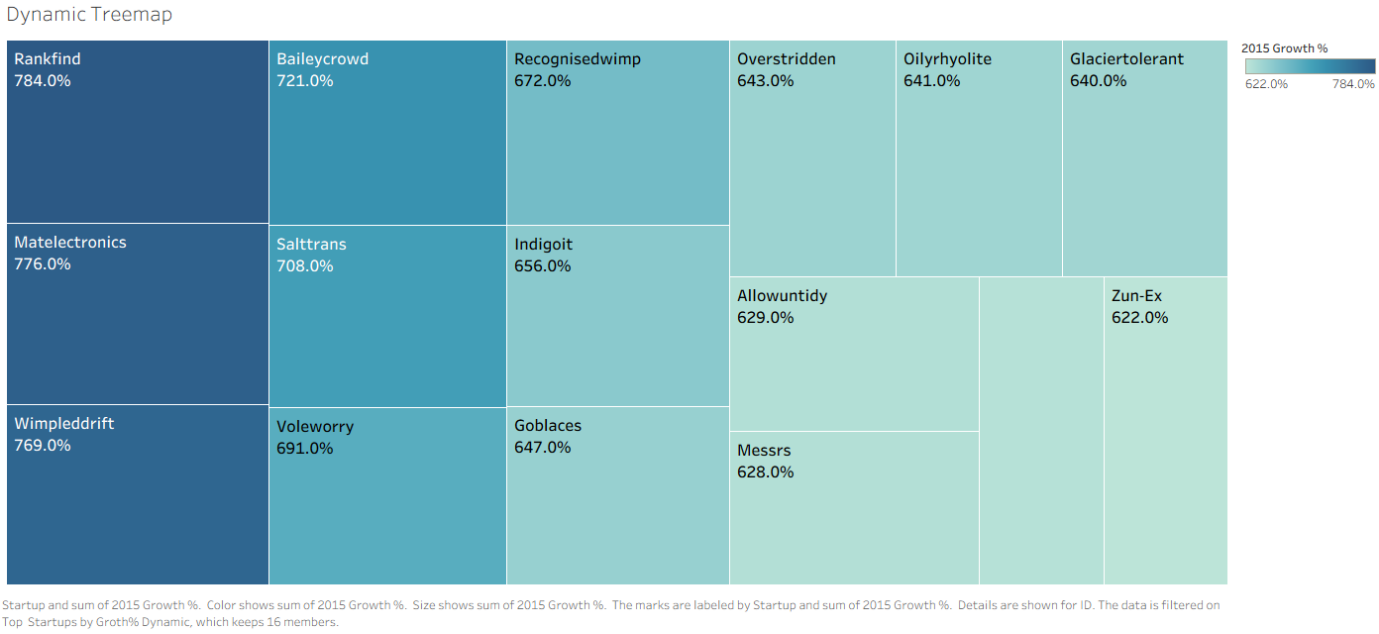


Industries -- Average Employees by Sector (Food & Beverage: 393.4, Energy: 330.6, Financial Services: 207.2)

Tableau technique: SUM([Employees])/COUNTD([Startup]) aggregated by [Industry], sorted descending. Reference line at grand average for quick benchmark comparison.

4 VIEW 2 -- DYNAMIC TREEMAP: TOP STARTUPS BY GROWTH %

A parameter-driven treemap where tile size and color both encode 2015 Growth %. A "Top Startups by Growth% Dynamic" set uses a parameter to dynamically filter to the top N performers. Color scale ranges from dark navy (highest: Rankfind 784%) to light teal (lowest in set: Zun-Ex 622%).

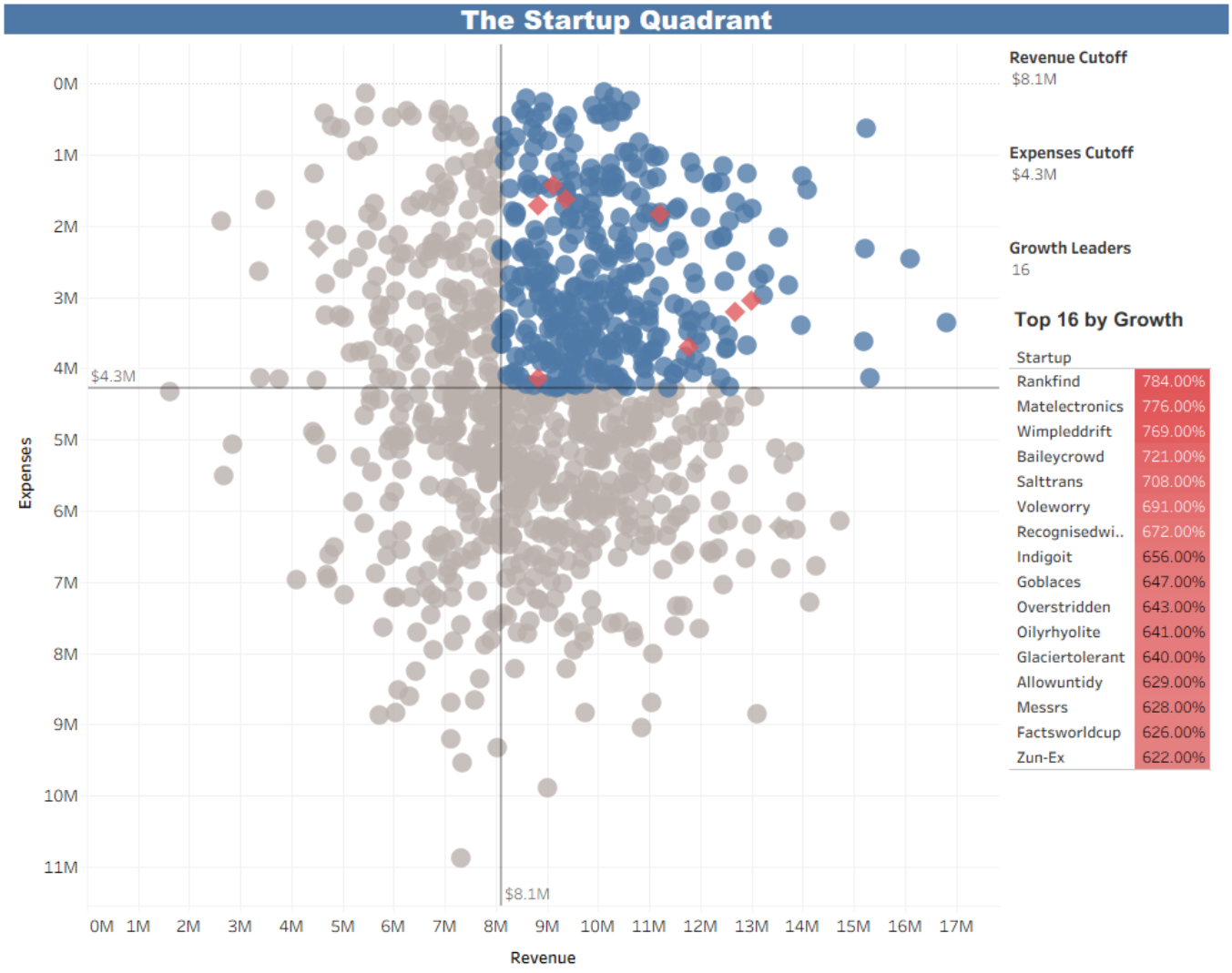


Dynamic Treemap -- Top 16 Startups by 2015 Growth %. Size and color both encode growth %. Rankfind leads at 784%.

> *Tableau technique: Set created with Top N condition using parameter [Growth Leaders]. SUM([2015 Growth %]) drives both size (Rows shelf) and color (Marks card). The data is filtered ON the set membership.*

5 VIEW 3 -- STARTUP QUADRANT: REVENUE vs EXPENSES

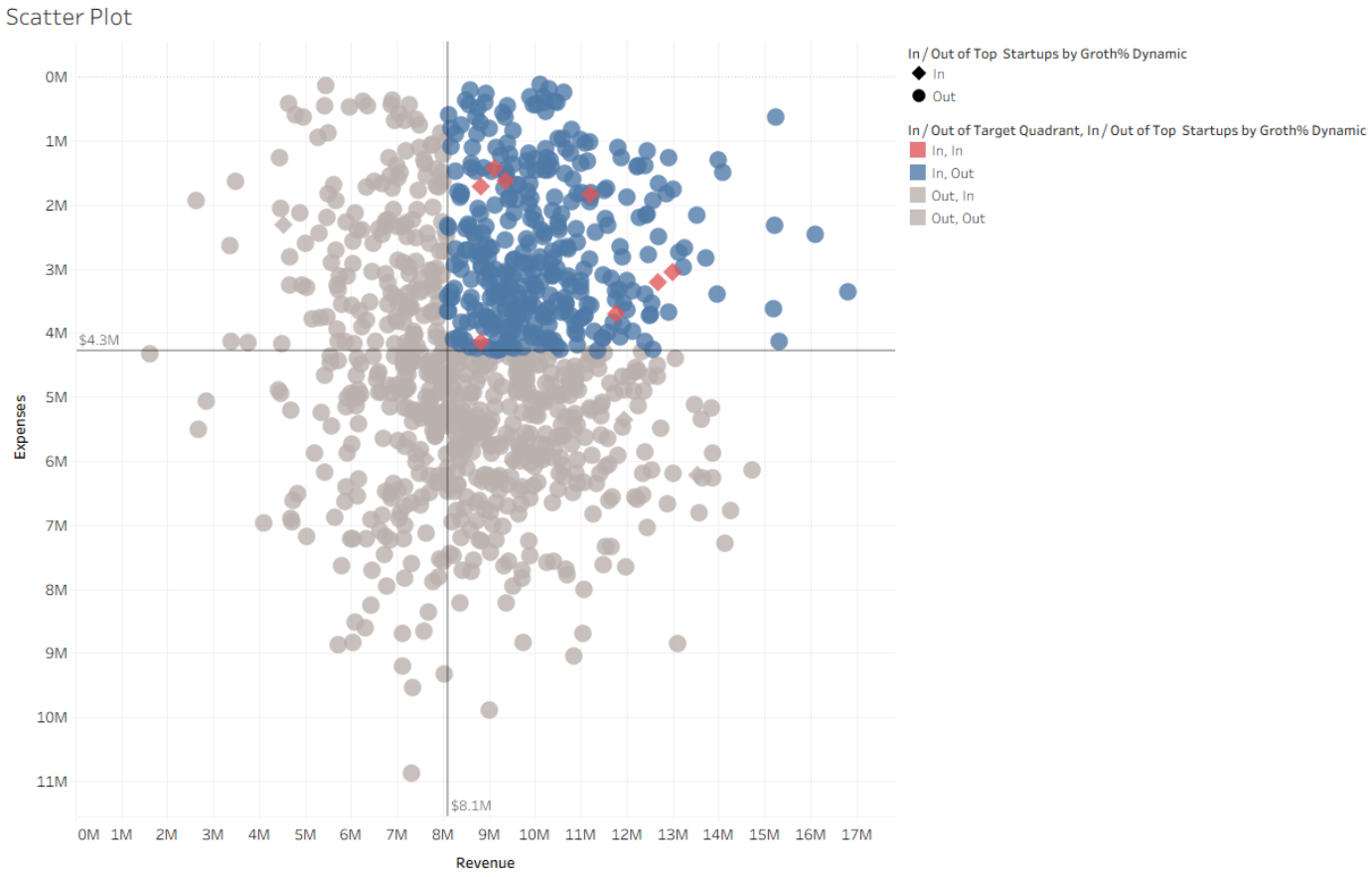
The primary analytical view. Each bubble represents a startup plotted on Revenue (x-axis) vs Expenses (y-axis). Two parameter-controlled reference lines at $8.1M Revenue and $4.3M Expenses divide the space into 4 quadrants. The upper-left quadrant (high revenue, low expenses) is the target zone, shown in blue. Top 16 growth leaders are highlighted as red diamonds.



Startup Quadrant -- Revenue ($8.1M cutoff) vs Expenses ($4.3M cutoff). Red diamonds = Top 16 growth leaders. Blue = in target quadrant.

6 VIEW 4 -- SCATTER PLOT: DUAL ENCODING WITH SHAPE + COLOR

An enhanced version of the quadrant scatter adding shape encoding: diamonds indicate membership in the Top Growth set; circles indicate non-membership. Color uses a 4-category scheme encoding the intersection of Target Quadrant (In/Out) and Top Startups by Growth (In/Out), creating four distinct visual groups for cross-dimensional analysis.



Scatter Plot -- Dual encoding: shape (diamond=In top growth, circle=Out) + 4-color quadrant/growth classification.

- In, In (red/salmon): In target quadrant AND in top growth set -- prime investment targets.
- In, Out (blue): In target quadrant but not in top growth set -- financially healthy but lower growth.
- Out, In (lighter grey): In top growth but outside target quadrant -- high growth but financially stretched.
- Out, Out (dark grey): Neither in target quadrant nor top growth -- lowest priority.

Tableau technique: Calculated fields for set membership combined via IF/ELSEIF logic. Shape shelf uses [In/Out of Top Startups by Growth%] set; Color shelf uses the combined 4-way classification field.

7 TABLEAU SKILLS DEMONSTRATED

- Parameters: Revenue Cutoff, Expenses Cutoff, Growth Leaders (N) -- all user-adjustable at runtime.
- Sets: Top N by Growth% Dynamic set with parameter-linked condition for dynamic treemap filtering.
- Reference Lines: Dual reference lines on scatter plot creating interactive quadrant boundaries.
- Dual mark encoding: Independent control of shape (set membership) and color (quadrant classification) on a single scatter.
- Treemap: SUM(growth%) driving both size and color with a 2-color diverging scale.
- Sorted horizontal bar chart with grand average reference line for industry benchmarking.
- Cross-view consistency: Same startup data supports 4 complementary analytical perspectives in one workbook.